

Web Application Vulnerability Scanner

Internship Project Report

Introduction

Web applications are widely used for services such as banking, shopping, education, and communication. Due to their public availability on the internet, they are common targets for cyber attacks. Many web applications suffer from security vulnerabilities because of improper input validation and weak security practices. This project focuses on building a basic Web Application Vulnerability Scanner to identify common security issues such as Cross-Site Scripting (XSS), SQL Injection (SQLi), and Cross-Site Request Forgery (CSRF).

Abstract

The Web Application Vulnerability Scanner is a Python-based tool designed to detect common vulnerabilities listed in the OWASP Top 10. The scanner crawls a target web application, identifies input fields and URLs, injects test payloads, and analyzes server responses to detect potential security flaws. A simple web interface is developed using Flask to allow users to manage scans and view results easily.

Tools Used

- Python
- requests
- BeautifulSoup
- Regular Expressions (Regex)
- Flask
- OWASP Top 10

Steps Involved in Building the Project

1. Crawling the target website to find URLs and input fields.
2. Injecting test payloads for XSS and SQL Injection.
3. Analyzing server responses for vulnerability indicators.
4. Detecting vulnerabilities using pattern matching and regex.
5. Creating a Flask-based web interface to manage scans.
6. Logging vulnerabilities with evidence and severity levels.

Conclusion

This project provided hands-on experience in understanding web application security and common vulnerabilities. By building a basic vulnerability scanner, I learned how automated tools identify security flaws and why secure coding practices are important. The project improved my practical knowledge of cyber security and vulnerability assessment.